

First three entries ($N \geq 8$);
$a_k = \widehat{f}_{2k}$, $b_k = \widehat{f}_{2k+1}$.

$$0 \leq n < N/2, \quad f_{e,n} = f_n + f_{n+N/2}, \quad f_{o,n} = f_n - f_{n+N/2}$$

$$\alpha_{N,n} = e^{-2\pi i n/N}, \qquad 0 \leq k < N/2$$